higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

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NATIONAL CERTIFICATE

MECHANOTECHNICS N5

(8190225)

10 April 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages, 1 information sheet and a formula sheet of 3 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MECHANOTECHNICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

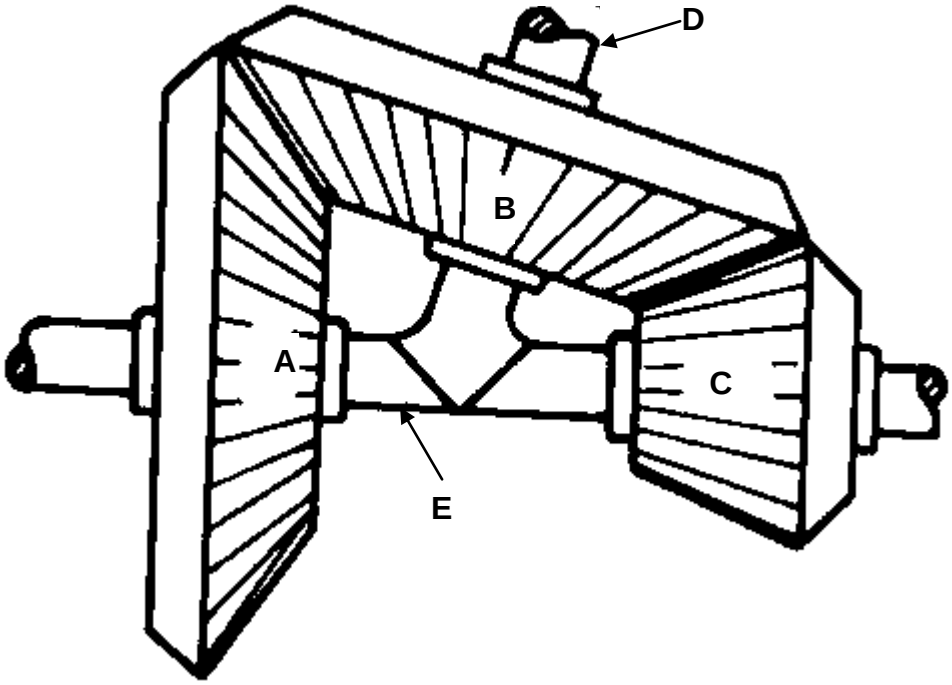
1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled
5. Write neatly and legibly.

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QUESTION 1

1.1 Name THREE types of gears that are utilised for reduction gearboxes. (3)

1.2 The FIGURE below shows an epicyclic gear train consisting of bevel gears and two arms (E and D).
Gear A has 40 teeth
Gear B has 32 teeth
Gear C has 24 teeth
E and D are two arms



Calculate the following:

1.2.1 The speed of gear B if wheel A is fixed and arms D and E rotate at 120 r/min in a clockwise direction (11)

1.2.2 The speed of gear C if wheel A is fixed and arms D and E rotate at 120 r/min in a clockwise direction (4)
[18]

QUESTION 2

2.1 Name FOUR examples of materials that can be transported by pipe conveyors. (4)

2.2 A bucket elevator is used to lift coal with a density of 900 kg/m^3 at a rate of 400 ton/h through a vertical height of 50 m. The chain speed is 0,4 m/s and the spacing between the buckets is 1 m.

Determine the following:

2.2.1 The volume of each bucket (8)

2.2.2 The required power of the driving motor if the head efficiency is 80% (5)

[17]

QUESTION 3

3.1 State FOUR advantages of aerial ropeways over other means of transport. (4)

3.2 A ropeway on a monocable system carries 460 kg loads at 90 m intervals, the span is 90 m and the sag with the load in the middle is 2,6 m. The ultimate strength of the cable is 600 kN and the stress should not exceed 1 300 MPa.

Calculate:

3.2.1 The diameter of the rope of a 6/7 construction (4)

3.2.2 The mass of the rope per metre (2)

3.2.3 The tension due to self-weight (2)

3.2.4 The tension due to one load (2)

3.2.5 The factor of safety (4)

[18]

QUESTION 4

4.1 State FOUR specifications of ropes. (4)

4.2 The cage of a goods elevator carries a total load of 1 600 kg. A rope passes over a drum with a diameter of 1 m and then to a balance a mass of 500 kg. The mass of rope including balance mass rope is 270 kg. The frictional torque on the drum shaft is 900 Nm.

If the maximum speed of the elevator is 6 m/s, calculate the power required to accelerate the elevator to a maximum speed in 8 s. (13)

[17]

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QUESTION 5

5.1 A locomotive travels at 40 km/h on a level track and accelerate uniformly to attain a speed of 100 km/h in 30 s. Ignore the rolling resistance between the wheels and the rail.

Calculate:

5.1.1 The initial and final velocity

5.1.2 The acceleration

(2 × 2) (4)

5.2 An electric train with a total mass of 350 tons is accelerating at 0,09 m/s on an incline of 1 in 130. The rolling resistance is 0,89% of the mass of the train.

Calculate:

5.2.1 The locomotive tractive effort required to overcome friction resistance (2)

5.2.2 The locomotive tractive effort required to overcome gravity (2)

5.2.3 The tractive effort of the locomotive required to accelerate the train (2)

5.2.4 The power required at a speed of 64 km/h (4)

5.2.5 The kinetic energy of the locomotive at 63 km/h (2)

[16]

QUESTION 6

6.1 A flywheel has a mass of 1 400 kg and a radius of gyration of 820 mm.

Calculate the torque required to increase the speed of the flywheel from 100 r/min to 130 r/min in 35 seconds. (8)

6.2 A flywheel has a mass of 600 kg and a radius of gyration of 830 mm.

Calculate the following:

6.2.1 The kinetic energy required to rotate the flywheel at a speed of 95 r/min

6.2.2 The uniform torque required to accelerate the flywheel from 48 r/min to 108 r/min

6.2.3 The time required for this acceleration

(3 × 2) (6)

[14]

TOTAL: 100

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INFORMATION SHEET

SUMMARY OF ROPE AREAS

NO.	TYPE OF ROPE	MASS/METRE LENGTH	ULTIMATE STRENGTH
1.	6/7 construction	$\frac{d^2}{250} = kg/m$	*Rope area* x ultimate stress $0{,}36D^2 \times \sigma$
2.	Locked coil type	$\frac{d^2}{175} = kg/m$	$0{,}6D^2 \times \sigma$
3.	Spiral rope type	$\frac{d^2}{195} = kg/m$	$0{,}55D^2 \times \sigma$
4.	Flexible type	$\frac{d^2}{250} = kg/m$	$0{,}31D^2 \times \sigma$

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MECHANOTECNICS N5

FORMULA SHEET

1. $m = \frac{PCD}{T}$
2. $DO = m \times (T + 2)$
3. $C = \frac{m}{2} \times (TA + TB)$
4. $Ke = \frac{1}{2} mv^2$
5. $VR = \frac{TA}{TB}$
6. $VR = \frac{PCD\ of\ gear}{PCD\ of\ pinion}$
7. $VR = \frac{NB}{NA}$
8. $NA \times TA = NB \times TB$
9. $Ft = \frac{2 \times T}{PCD}$
10. $Fr = Ft \times Tan\phi$
11. $Fn = Ft \times Sec\phi$
12. $Ie = IA + (VR)^2\ IB + (VR)^2\ IC + (VR)^2\ ID$
13. $T\alpha = Ie \times \alpha A$
14. $T\alpha = TA + \frac{(NB)}{(NA)} \frac{TBC}{\eta 1} + \frac{(ND)}{(NA)} \frac{TD}{\eta 1 \eta 2}$
15. $\frac{NB}{NA} = \frac{\omega B}{\omega A} = \frac{\alpha B}{\alpha A} = \frac{IA}{IB}$
16. $T_{OUTPUT} = T_{INPUT} \times GR \times \eta$
17. $P = \frac{\pi \times PCD}{n}$
18. $Ti + To + Th = 0$
19. $TA = TS + 2TP$
20. $\frac{Input\ speed}{Output\ speed} = \frac{Teeth\ on\ driven\ gears}{Teeth\ on\ driving\ gears}$
21. $v = \pi \times (d + t) \times N$
22. $P = Te \times v$
23. $\frac{T1}{T2} = e^{\mu\theta}$
24. $T1 = \delta \times A$
25. $Tc = m \times v^2$
26. $\frac{T1 - TC}{T2 - TC} = e^{\mu\theta cosec\ \alpha}$
27. $L = \frac{\pi}{2} \times (D + d) + \frac{(D \pm d)^2}{4 \times C} + 2C$
28. $Tg = m \times g \times sin\phi$
29. $v = \omega \times r$

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30. $v = \sqrt{\mu \times g \times r}$

31. $v = \sqrt{\frac{g \times b \times r}{2 \times h}}$

32. $v = \sqrt{gr\left[\frac{\mu + Tan\,\theta}{1 - \mu Tan\,\theta}\right]}$

33. $v = \sqrt{gr\left[\frac{hTan\theta + b/2}{h - b/2\,tan\,\theta}\right]}$

34. $\frac{T1}{T2} = \left[\frac{1 + \mu Tan\,\theta}{1 - \mu Tan\,\theta}\right]^n$

35. $Cos\,\frac{\theta}{2} = \frac{R - r}{C}$

36. $Cos\,\frac{\phi}{2} = \frac{R + r}{C}$

37. $m = w \times t \times L \times \rho$

38. $T1 = w \times n \times ft$

39. $P = Pg + P\mu$

40. $t = \frac{I \times \omega}{T}$

41. $P = \frac{2 \times \pi \times N \times T}{60}$

42. $T = F \times r$

43. $w = do + 3d - 1{,}5155P$

44. $do = de + 0{,}65P$

45. $w = \frac{\pi \times m}{2}(cos^2\,\theta)$

46. $h = m\left[1 - \frac{\pi}{4}\left(sin\,\theta\ Cos\theta\right)\right]$

47. $\frac{p1}{Rho} + \frac{(v1)^2}{2} + gh1 = \frac{p2}{Rho} + \frac{(v2)^2}{2} + gh2$

48. $Vw\,(Va) = \sqrt{\frac{gx^2}{2y}}$

49. $v = C\sqrt{mi}$

50. $hf = \frac{4 \times f \times \ell \times v^2}{2 \times g \times d}$

51. $hf = \frac{f \times \ell \times O^2}{3{,}026 \times d^5}$

52. $Q = \frac{Cd \times A \times a \times \sqrt{(2gh)}}{\sqrt{(A^2 - a^2)}}$

53. $Q = Cd \times A \times \frac{\sqrt{(2gh)}}{\sqrt{(m^2 - 1)}}$

54. $V = \sqrt{(g \times R \times Cos\theta)}$

55. Vol. bucket = $\frac{m \times s}{\rho \times v}$

56. $L = 2C + \pi D$

57. Self-weight = $\frac{m1 \times g \times S^2}{8 \times h}$

58. One load = $\frac{m2 \times g \times S}{4 \times h}$

59. T (acc load) = (T1 - T2)R

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60. T (acc drum) = $I \times \alpha = mk^2 \times \frac{a}{R}$

61. P = ω × T

62. ω = 2π × N

63. $Ke = \frac{1}{2} I \times \omega^2$

64. $Ke = \frac{work\ done}{efficiency}$

65. P = Ke × operations/second

66. (I_1 + I_2) ω_3 = $I_1\omega_1$ + $I_2\omega_2$

67. μ = Tan θ

68. $\eta = \frac{Tan\,\theta}{Tan\,(\theta + \phi)}$

69. T = μ × F × Re × n

70. $T = \frac{\mu \times F \times Re}{sin\,\theta}$

71. T = μ × n × (Fc - S)R

72. Fc = m × ω^2 × γ

73. $\mathrm{Fc} = \frac{\mathrm{mv}^2}{\gamma}$

74. Tractive effort = mass on driving wheels × μ × g

75. Side thrust = $FcCos\theta - mg\ Sin\theta$

76. $\mu = \frac{FcCos\theta - mgSin\theta}{mgCos\theta + FcSin\theta}$

77. $P_l = CmgL + mgh$

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